Introduction

A z-score expresses each observation as its distance from the mean in units of the standard deviation. Standardisation is often applied before clustering, PCA, ridge/lasso regression, and any procedure whose output is unit-sensitive. For data with outliers, a robust variant using the median and MAD is safer.

Prerequisites

The reader should know what the mean, SD, median, and MAD are.

Theory

The classical z-score for observation x_i is

$$z_i = (x_i - \bar{x})/s.$$

After standardisation, the sample mean is 0 and the sample SD is 1. The values are dimensionless and comparable across variables measured on different scales.

The **robust z** uses the median and scaled MAD:

$$z_i^{\text{robust}} = (x_i - \text{median}(x))/(1.4826 \cdot \text{MAD}(x)).$$

It reduces to the classical z for normal data but resists the inflation of the scale estimator by outliers.

Standardisation is not the same as normalisation. **Min-max normalisation** rescales to $[0, 1]$: $(x_i - \min(x))/(\max(x) - \min(x))$. Each has its use; z-scores preserve relative distances better.

Assumptions

Purely algebraic; no distributional assumption. The choice between classical and robust z depends on whether outliers are present.

R Implementation

```
library(robustbase)

set.seed(2026)
x <- c(rnorm(50, mean = 100, sd = 10), 200, 250)

z_classical <- scale(x)
z_robust    <- (x - median(x)) / (1.4826 * mad(x, constant = 1))

range(z_classical)
range(z_robust)

cohort <- data.frame(
  age   = rnorm(100, 58, 12),
  sbp   = rnorm(100, 130, 18),
  crp   = rlnorm(100, log(3), 0.6)
)
z_cohort <- scale(cohort)
colMeans(z_cohort)
apply(z_cohort, 2, sd)
```

Output & Results

```
range(z_classical): -1.69   6.11
range(z_robust):    -2.02  15.04
```

The robust z flags the outlier at 250 as > 15 SDs from the median, while the classical z only gives 6. Classical z understates the deviation because the extreme point has inflated the sample SD.

Interpretation

Report z-scores when comparing observations on different scales. Biological examples: centred and scaled gene-expression values across samples; cognitive-test scores scaled to a reference population.

Practical Tips

- `scale()` returns a matrix; wrap with `as.data.frame()` when using downstream tidyverse functions.
- Save the `center` and `scale` attributes to apply the same standardisation to new test data – critical in predictive modelling.
- For skewed variables, log-transform before standardising.
- The robust z is a simple and effective first-pass outlier filter: $|z^{\text{robust}}| > 3$ flags candidates.
- Do not standardise the outcome in a regression if interpretability is desired; standardise predictors if comparing coefficients.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1